25901340 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCICH200 - INSTRUMENTATION TECHNIQUES IN CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries **1** marks

1. The time constant of an instrument indicates [A] / [CO1]

   a). Sensitivity b). Speed of response

   c). Accuracy d). Linearity

2. Float type level indicators operate based on [U] / [CO1]

   a). Capacitance b). Buoyancy c). Voltage d). Radiation

3. Which of the following is an example of chemical noise? [U] / [CO1]

   a). Thermal noise b). Shot noise

   c). Reagent impurities d). Electronic drift

4. Which type of spectroscopy involves transitions between electronic energy levels? [K] / [CO2, CO3]

a). Microwave spectroscopy b). Infrared spectroscopy

c). UV–Visible spectroscopy d). NMR spectroscopy

5. Which device disperses light into its component wavelengths and allows selection of a narrow wavelength band? [K] / [CO2]

a). Detector b). Monochromator

c). Amplifier d). Cuvette

6. The signal produced by a detector is usually proportional to: [U] / [CO2]

a). Sample weight b). Light intensity

c). Sample concentration d). Temperature

7. What is the unit of specific rotation in polarimetry? [U] / [CO4]

a). Gram/milliliter b). Degrees Celsius

c). Gram/cm^3 d). Degrees/(g/mL)

8. In a polarimeter, what is the primary function of a polarizer? [A] / [CO4]

a). To detect the interference patterns of light passing through a sample.

b). To convert polarized light into unpolarized light.

c). To analyze the color changes in the sample.

d). To select and transmit light of a specific polarization.

9. While measuring the refractive index of a volatile liquid, the reading fluctuates continuously. Which instrument component should be checked first? [A] / [CO4]

a). Eyepiece b). Water circulation system

c). Mirror d). Scale

10. Beer–Lambert law is applied in UV–Visible spectroscopy to determine: [U] / [CO3]

a). Molecular weight b). Concentration of solution

c). pH of solution d). Boiling point

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **3** marks

11.	Explain how automatic instruments reduce human error.	[A] / [CO1]
12.	What is time constant? Explain its importance in first-order instruments.	[A] / [CO1]
13.	Calculate the energy of radiation having wavelength 400 nm. ($h = 6.626 \times 10^{-34}$ J·s, $c = 3 \times 10^{8}$ m/s).	[A] / [CO2, CO3]
14.	How is IR spectroscopy used to identify functional groups?	[U] / [CO3]
15.	State Snell's law and write its mathematical expression.	[U] / [CO4]
16.	What is the function of the reference material holder?	[U] / [CO4]

[3x4 = 12]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **7** marks

17.	Discuss the significance of S/N ratio in analytical techniques such as spectroscopy and chromatography.	[U] / [CO1]
18.	What is a flicker noise? What are its characteristics? What are the methods to reduce it?	[U] / [CO1]
19.	Write short notes on: (a) Nernst Glower (b) Globar Source.	[U] / [CO2]
20.	Derive and explain the Beer–Lambert law.	[U] / [CO3]
21.	Define Thermal Gravimetric Analysis (TGA). Explain its principle, instrumentation and working in detail.	[E] / [CO4]
22.	Write a note on different sampling methods.	[U] / [CO3]

[7x4 = 28]